

1. What is a conditional statement in programming?

Ans. A conditional statement is used to make decisions based on whether a condition is true or false.

2. What does an if statement do?

Ans. It executes a block of code if a specified condition is true.

3. What is the purpose of an else block?

Ans. It executes code when the if condition is false.

4. When is an else block executed?

Ans. When the condition in the if statement evaluates to false.

5. What is a Boolean expression?

Ans. An expression that evaluates to either true or false.

6. Write the syntax of a simple if statement.

Ans. `if(condition)`

```
{  
    // statements  
}
```

7. What operator is commonly used to test equality?

Ans. `==`

8. What does the modulus (%) operator return?

Ans. Modulus operator (%) returns the remainder after division.

9. How can you check if a number is even?

Ans. `num % 2 == 0`

10. How can you check if a number is odd?

Ans. `num % 2 != 0`

11. What is the purpose of else-if?

Ans. To test multiple conditions sequentially.

12. What is a nested if statement?

Ans. An if statement inside another if statement.

13. Can an if statement exist without an else?

Ans. Yes.

14. What is the output when a false condition is tested in an if statement without else?

Ans. No output; the code inside if is skipped.

15. What is the role of braces {} in C/C++ conditionals?

Ans. They group multiple statements into a block.

16. What does > mean?

Ans. Greater than.

17. What does < mean?

Ans. Less than.

18. What does >= mean?

Ans. Greater than or equal to.

19. What does <= mean?

Ans. Less than or equal to.

20. What does != mean?

Ans. Not equal to.

21. What is a switch statement?

Ans. A decision-making statement that selects one block from many choices.

22. What is a case label in a switch statement?

Ans. A value that identifies a specific block of code.

23. Why is break used in switch?

Ans. To exit the switch after executing a case.

24. What happens if break is omitted?

Ans. Execution continues to the next case (fall-through).

25. What is the default case in switch?

Ans. The block executed when no case matches.

26. Can switch compare strings in C++?

Ans. No, traditional switch works with integral values, not strings.

27. Which statement is better for many constant choices: if-else or switch?

Ans. Switch statement.

28. How do you test whether a value is positive?

Ans. `value > 0`

29. How do you test whether a value is negative?

Ans. `value < 0`

30. How do you test whether a value is zero?

Ans. `value == 0`

31. Write a condition to check if age is at least 18.

Ans. `age >= 18`

32. Write a condition to check if marks are above 50.

Ans. `marks > 50`

33. What is decision making in programming?

Ans. The process of choosing actions based on conditions.

34. Can nested if statements have multiple levels?

Ans. Yes.

35. What is logical AND (&)?

Ans. Returns true only if all conditions are true.

36. What is logical OR (||)?

Ans. Returns true if at least one condition is true.

37. What is logical NOT (!)?

Ans. Reverses the logical value of a condition.

38. How do you check if a year is divisible by 4?

Ans. `year % 4 == 0`

39. What is a leap year?

Ans. A year with 366 days instead of 365.

40. What is the first condition for a leap year?

Ans. The year must be divisible by 4.

41. How do you find the largest of two numbers?

Ans. if($a > b$)

then a is larger; otherwise b is larger.

42. How do you find the largest of three numbers?

Ans. Use if-else statements to compare all three numbers.

43. What is an if-else-if ladder?

Ans. A sequence of if, else if, and else statements used to test multiple conditions.

44. What is the difference between if and switch?

Ans. if can test any condition, while switch checks a variable against constant values.

45. Can switch use ranges directly?

Ans. No.

46. What is fall-through in switch?

Ans. Execution moving to the next case when break is omitted.

47. What data type is usually used for conditions?

Ans. bool (Boolean).

48. Why are conditional statements important?

Ans. They allow programs to make decisions and perform different actions.

49. Give one real-life example of a conditional statement.

Ans. If the traffic light is green, go; otherwise, stop.

50. Name four conditional structures covered.

Ans.

1. if
2. if-else
3. else-if ladder
4. switch statement